SOFTWARE REQUIREMENTS SPECIFICATION

Movie Information Portal

Web Technology – I (2301CS363)

Project Duration: 8 Weeks
Project Type: Web-Based Frontend Application

Technology Sequence

HTML5 → CSS3 → JavaScript → DOM → jQuery → Bootstrap

1. Introduction

1.1 Purpose

The purpose of this project is to develop a Movie Information Portal, a web-based application that allows users to browse, search, filter, and view information about movies. The project provides practical experience in developing a complete web application using the technologies covered in Web Technology–I.
The project will progressively use HTML, CSS, JavaScript, DOM, jQuery and Bootstrap.

1.2 Project Overview

The Movie Information Portal provides users with information about movies and interactive features such as searching, filtering, sorting, favorites, ratings, reviews and feedback.

1.3 Project Title

Movie Information Portal

2. Objectives

- Develop a structured web application using HTML5.
- Apply CSS3 concepts for webpage design and layout.
- Implement programming logic using JavaScript.
- Dynamically manipulate webpage content using the DOM.
- Use JavaScript events for user interaction.
- Implement client-side form validation.
- Use jQuery for interactive webpage effects and events.
- Use Bootstrap for responsive webpage design and UI components.
- Integrate multiple Web Technology concepts into one practical project.
- Develop a complete mini-project based on the Web Technology–I syllabus.

3. Scope of the Project

The project focuses on frontend development of a Movie Information Portal. The application will display movie information such as title, poster, genre, release year/date, language, duration, director, cast, description, rating and trailer information.

A backend server or database is not mandatory. Students may maintain movie information using JavaScript arrays and objects. LocalStorage may be used as an optional advanced feature.

4. Target Users

4.1 Movie Viewer

- Browse movies.
- Search for movies.
- Filter and sort movies.
- View movie details.
- Add movies to favorites/watchlist.
- Submit ratings and reviews.

4.2 Project Administrator – Optional

- Add movies.
- Edit movie information.
- Delete movie information.

The administrator feature is optional and is not mandatory for the basic project.

5. Project Modules

Module	Main Features
1. Home	Banner, featured movies, popular movies, genres and navigation.
2. Movie Listing	Movie cards, poster, title, genre, year and rating.
3. Movie Details	Poster, description, cast, director, language, duration, rating and trailer.
4. Search & Filter	Search by title/keyword and filter by genre, language, year and rating.
5. Favorites / Watchlist	Add and remove movies from favorites and display the favorite list.
6. Reviews & Ratings	Rating, review submission and dynamic review display.
7. Contact / Feedback	Contact form, feedback form and client-side validation.
8. Advanced Feature	Recommendation, dark/light mode, comparison, LocalStorage or another approved feature.

6. Proposed Website Pages

- index.html – Home
- movies.html – Movie Listing
- movie-details.html – Movie Details
- favorites.html – Favorites / Watchlist
- reviews.html – Reviews & Ratings
- contact.html – Contact / Feedback

7. Functional Requirements

FR-01: Home Page

- Website logo/title.
- Navigation menu.
- Movie banner.
- Featured movies.
- Popular movies.
- Movie genres.
- Footer.

FR-02: Display Movies

- Display a collection of movies.
- Show movie poster, title, genre, release year and rating.
- Provide a short description and View Details action.
- Use consistent movie-card design.

FR-03: View Movie Details

- Display title, poster, description, genre, release year, language, duration, director, cast, rating and trailer.

FR-04: Search Movies

- Search by movie title or keyword.
- Display search results dynamically.
- Display an appropriate no-results message.

FR-05: Filter Movies

- Filter by genre.
- Filter by language.

- Filter by release year.
- Filter by rating.

FR-06: Sort Movies

- Sort by movie name A–Z or Z–A.
- Sort by rating high-to-low or low-to-high.
- Sort by release year newest-to-oldest or oldest-to-newest.

FR-07: Add Movie to Favorites

- Allow users to add a movie to favorites/watchlist.
- Display selected movies on the Favorites page.

FR-08: Remove Movie from Favorites

- Allow users to remove a movie from the favorites list.

FR-09: Submit Movie Review

- Provide a review form containing user name, email, movie, rating and review.

FR-10: Movie Rating

- Allow users to provide a movie rating.
- Display the selected rating appropriately.

FR-11: Display Reviews

- Display submitted reviews dynamically.
- Show user name, rating, review and movie name where appropriate.

FR-12: Contact Form

- Provide name, email, subject, message and submit controls.

FR-13: Form Validation

- Validate required fields.
- Validate email format.
- Validate minimum/maximum input length where appropriate.
- Validate rating range.
- Display meaningful error messages.

8. User Interface Requirements

UI-01: Navigation

Provide consistent navigation across pages. Suggested navigation: Home | Movies | Favorites | Reviews | Contact.

UI-02: Movie Cards

Each card should contain poster, title, genre, rating, release year and an action button.

UI-03: Responsive Design

The website should support desktop, laptop, tablet and mobile screen sizes.

UI-04: Visual Design

Maintain consistent typography, colors, spacing, alignment, buttons, icons and navigation.

9. Suggested Project Structure

```
MovieInformationPortal/
│
├── index.html
├── movies.html
├── movie-details.html
├── favorites.html
├── reviews.html
├── contact.html
│
├── css/
│   └── style.css
│
├── js/
│   ├── movies.js
│   ├── app.js
│   ├── favorites.js
│   └── validation.js
│
├── images/
│   ├── posters/
│   └── banners/
│
└── README.txt
```

10. Movie Data Requirements

Students should maintain a minimum collection of 15 movies.

- Movie ID
- Movie Title
- Poster
- Genre
- Release Year
- Language
- Duration
- Director
- Cast
- Description
- Rating
- Trailer

```
Example JavaScript movie object:
{
  id: 1,
  title: "Inception",
  genre: "Sci-Fi",
  year: 2010,
  language: "English",
  duration: "2h 28m",
  director: "Christopher Nolan",
  rating: 8.8,
  description: "A science-fiction thriller..."
}
```

11. Non-Functional Requirements

NFR-01: Usability

The website should be simple and easy to navigate.

NFR-02: Responsiveness

The website should work properly on different screen sizes.

NFR-03: Performance

The webpage should load reasonably quickly and avoid unnecessary large resources.

NFR-04: Consistency

All pages should maintain a consistent visual design.

NFR-05: Maintainability

HTML, CSS and JavaScript should preferably be separated into appropriate files.

NFR-06: Browser Compatibility

The application should work in modern browsers such as Google Chrome, Microsoft Edge and Mozilla Firefox.

12. Project Development Schedule

Week	Date	Technology / Unit	Major Work	Expected Output
1	24/08/2026 to 30/08/2026	HTML	Project structure, Home, Movies, Details, forms.	Static HTML Portal
2	31/08/2026 to 06/09/2026	CSS	Styling, layouts, movie cards and CSS3.	Designed Portal
3	07/09/2026 to 13/09/2026	HTML + CSS	Remaining pages, responsive CSS and complete UI.	Complete Static UI
	14/09/2026 to 19/09/2026	Phase-1 Evaluation	• All major pages completed. • Navigation implemented. • Movie cards and movie details created. • Forms created. • CSS design completed. • Responsive design implemented.	Complete static UI
4	14/09/2026 to 20/09/2026	JavaScript	Movie data, arrays, objects, functions and validation.	Basic Functionality
5	21/09/2026 to 27/09/2026	JavaScript + DOM	Dynamic cards, search, filter, sorting and details.	Dynamic Portal
6	28/09/2026 to 04/10/2026	JavaScript + DOM	Favorites, reviews, ratings and validation.	Complete Functional Portal
	05/10/2026 to 10/10/2026	Phase-2 Evaluation	• Phase-1 content • Movie data implemented. • Dynamic movie cards. • Search, filtering and sorting. • Dynamic movie details. • Favorites. • Reviews and ratings. • Form validation.	Complete functional portal.
7	05/10/2026 to 11/10/2026	jQuery	Events, effects and interactive UI.	Enhanced Portal
8	12/10/2026 to 18/10/2026	Bootstrap	Grid, cards, navbar, forms, components and advanced feature.	Final Project
	19/10/2026 to 31/10/2026	Phase-3 Evaluation	• Phase-1, Phase-2 content • jQuery interactions and effects. • Bootstrap grid, cards, navbar and forms. • Bootstrap components. • Responsive UI. • All modules integrated. • At least one advanced feature.	Complete functional portal.

13. Detailed Weekly Project Work

Week 1 – HTML: Project Structure

- Finalize project title, problem statement, objectives, target users, modules, pages and navigation.
- Create the project folder structure.
- Develop the Home page with header, logo, navigation, banner, featured/popular movies, genres and footer.
- Develop the Movie Listing page with poster, title, genre, release year, rating, description and View Details button.
- Develop the Movie Details page with poster, title, description, genre, director, cast, year, duration, language, rating and trailer.
- Create Review and Contact/Feedback forms using HTML5.

Week 2 – CSS: Page Design and Layout

- Create external style.css and apply it to all pages.
- Apply fonts, colors, backgrounds, margins, padding, borders and box-model properties.
- Style the navigation menu, movie cards, buttons and movie details layout.
- Use CSS positioning, pseudo-classes, pseudo-elements, transitions, animations, gradients, image styling and tooltips.

Week 3 – HTML + CSS: Complete UI and Responsive Design

- Complete Home, Movies, Movie Details, Favorites, Reviews and Contact pages.
- Implement responsive CSS using media queries.
- Design movie layouts for desktop, tablet and mobile.
- Style all forms and controls.
- Maintain consistent navigation, typography, cards, buttons and footer across pages.
- Add IDs/classes that will be used later by JavaScript and DOM.

Week 4 – JavaScript: Movie Data and Basic Functionality

- Create movie data using JavaScript arrays and objects.
- Implement variables, functions, conditions, loops and built-in functions.
- Create functions for displaying, finding, searching, filtering and sorting movies.
- Implement basic search logic.
- Implement client-side form validation.
- Implement JavaScript events such as click, input, change and submit.

Week 5 – DOM: Dynamic Movie Portal

- Select HTML elements using DOM methods.
- Generate movie cards dynamically from the movie array.
- Implement dynamic movie search.
- Implement genre, language, year and rating filters.
- Implement sorting by name, rating and release year.
- Display selected movie details dynamically.
- Display an appropriate message when no movie matches the criteria.

Week 6 – JavaScript + DOM: Favorites, Reviews and Validation

- Implement Add to Favorites and Remove from Favorites.
- Display favorite movies dynamically.
- Create review and rating functionality.
- Display submitted reviews dynamically using DOM.
- Calculate or display movie ratings as appropriate.
- Validate required fields, email, rating and review length.
- Use suitable ES6 features such as let, const, arrow functions and template literals.

Week 7 – jQuery: Interactivity and UI Enhancement

- Integrate jQuery.
- Use jQuery selectors and event handling.
- Implement show/hide, fade and slide effects.
- Add interactive effects to movie cards.
- Animate search/filter results where appropriate.
- Improve form interaction and success/error messages using jQuery.

Week 8 – Bootstrap: Responsive and Professional UI

- Integrate Bootstrap.
- Use containers, grid system and responsive breakpoints.
- Convert movie listings into responsive Bootstrap cards.
- Create a responsive Bootstrap navbar.
- Improve search, review and contact forms using Bootstrap.
- Use Bootstrap carousel, modal, alert, badge and button components where appropriate.
- Add Font Awesome icons.
- Implement at least one advanced feature: recommendation, dark/light mode, comparison, LocalStorage, advanced search or admin movie management.